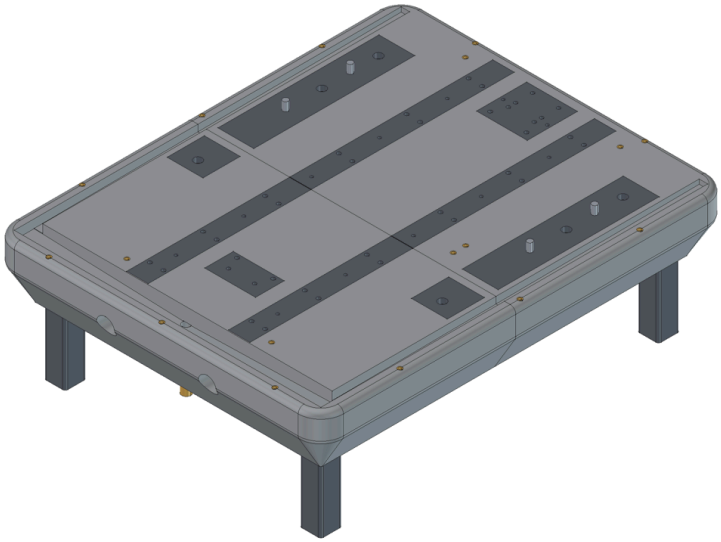
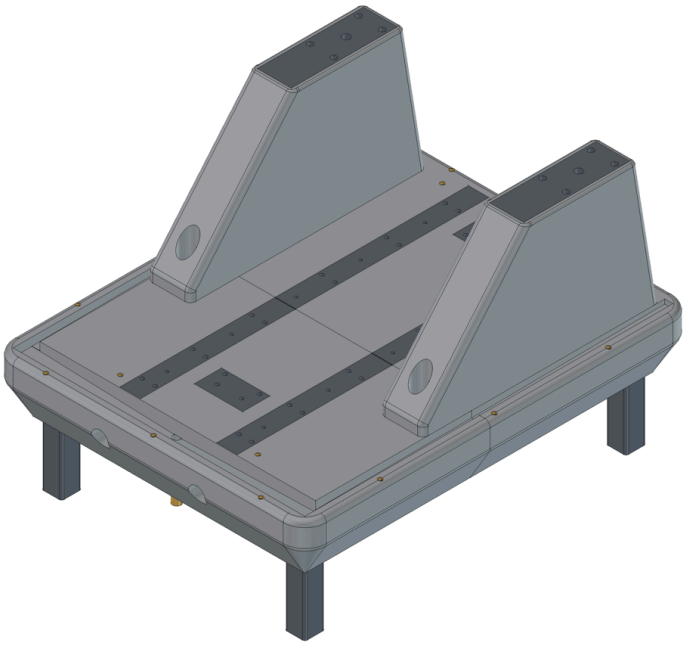
Build guide 3: frame and rail risers

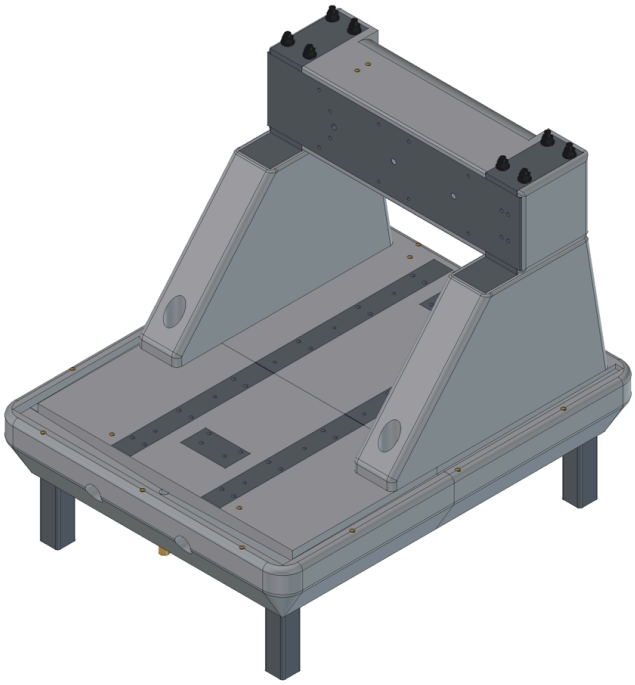
DIY CNC epoxy-granite mill · October 2026

4. Assemble the frame

1. **Feet:** set the rear base half on the bench on its legs. Bond a 1/2-13 nut into a printed plug in each leg end (or use the threaded inserts), screw in the leveling feet, and roughly level it.
2. **Join the base halves:** glue the O-ring cord into the rear half's split groove (glue the cord ends). Push the front half on, its split end plates over the rear half's dowel pins. Run the two 687 mm draw rods in from the front pockets, through both halves, a washer and nut on each end (the rod's end flush with the front nut), and tighten from the back pockets (snug, then about 40 N·m). The halves now meet end plate to end plate. Level the whole base with the feet.
3. **Grout the dowels.** The two dowel holes in each upright's foot are 1/2 in, over the 3/8 in dowels, so the fit comes from epoxy, not from the two molds agreeing. Wax the base dowels (three coats, buffed). Mix a little of the leftover epoxy with whiting to a thick cream and half-fill each dowel hole. Glue the foot O-ring into each upright's groove, lower the upright on (dowels into their holes, foot on the base) and put the tie rods in (next step) snug, wipe off what squeezes out, and leave it 24 h. After that the upright lifts off and drops back on in the same place.
4. **Tie rods.** Each upright is held by three 1/2-13 rods. Two run inside its columns and screw into the top plate's lower bar; one runs from the toe pocket, where its nut bears on the foot bar. All three go right through the base pedestal (inside its tube spacers) to its bottom bar. So the beam studs, the top plate, the columns, the foot bar, the pedestal and the rods are one steel chain, and every nut can be reached and retightened.
    - **Column rods:** push the 363 mm rod up from under the base, through the pedestal and the column, and screw it into the top plate's lower bar until it stops against the upper bar (finger tight).
    - **Toe rod:** washer and nut in the toe pocket; push the 153 mm rod up from below and screw it into the nut until its end is flush with the nut's top.
    - **Underneath:** an SAE washer and a nut on each rod against the pedestal's bottom bar. Snug all three, then tighten to about 40-50 N·m, front to back. Push the printed cap into the toe pocket, flush with the face (a dab of silicone holds it).
    - Check them again after a week and after the first month; later, once a year.
5. **Beam:** the top plates' dowel holes are 14 mm, oversize the same way. Screw the eight 190 mm 3/8-16 studs down through the top plates' upper bars into the lower bars until they stop. Wax the beam's dowels, put grout in the holes, set the beam down over the studs and run a washer and nut on each, snug. Check the beam face is square to the base top with a square and a dial indicator (shim under one top plate if needed) and leave it 24 h before the final torque (about 40 N·m).







5. Bed the rail risers

The risers are bonded to the castings with a thin epoxy grout and held down by screws into the steel: the base's rail bars and the beam's face bar, tapped M5.

1. Abrade and degrease the casting under each riser and the riser's underside.
2. Set the riser on small shim stacks (feeler stock) at each hold-down. Fit the M5 low-head screws (M5 x 25 for Y, M5 x 20 for X) just snug.
3. **Y risers (base):** make each one straight along its length and both coplanar with each other, within 0.02 mm over the length if you can. Check with the straightedge and feelers, or a dial indicator on a bridge spanning both. Set them parallel, 180 mm apart on center (rail spacing), using the rails as a gauge.
4. **X risers (beam face):** same, 100 mm apart. Make them parallel to the base top, checked with the indicator on the Y-axis table later, or now with a height gauge from the base top.
5. Dam the ends and sides with tape or plasticine, and flow low-viscosity epoxy thickened with a little whiting underneath from one side until it comes out the other.
6. After 48 h, torque the hold-downs to about 5 N·m (they're in steel) and recheck flatness.

